Chapter 1

Preliminaries

1.1 Introduction

In this work we describe how to train a multi-layer generative model of natural images. We use a dataset of millions of tiny colour images, described in the next section. This has been attempted by several groups but without success[3, 7]. The models on which we focus are RBMs (Restricted Boltzmann Machines) and DBNs (Deep Belief Networks). These models learn interesting-looking filters, which we show are more useful to a classifier than the raw pixels. We train the classifier on a labeled subset that we have collected and call the CIFAR-10 dataset.

1.2 Natural images

1.2.1 The dataset

The tiny images dataset on which we based all of our experiments was collected by colleagues at MIT and NYU over the span of six months; it is described in detail in [14]. They assembled it by searching the web for images of every non-abstract English noun in the lexical database WordNet[15, 8]. They used several search engines, including Google, Flickr, and Altavista and kept roughly the first 3000 results for each search term. After collecting all the images for a particular search term, they removed perfect duplicates and images in which an excessively large portion of the pixels were white, as they tended to be synthetic figures rather than natural images. The search term used to find an image provides it with a rough label, although it is extremely unreliable due to the nature of online image search technology.

In total, the dataset contains 80 million colour images downscaled to 32×32 and spread out across 79000 search terms. Most of our experiments with unsupervised learning were performed on a subset of about 2 million images.

1.2.2 Properties

Real images have various properties that other datasets do not. Many of these properties can be made apparent by examining the covariance matrix of the tiny images dataset. Figures 1.1 and 1.2 show this covariance matrix.

The main diagonal and its neighbours demonstrate the most apparent feature of real images: pixels are strongly correlated to nearby pixels and weakly correlated to faraway pixels. Various other properties can be observed as well, for example:

- The green value of one pixel is highly correlated to the green value of a neighbouring pixel, but slightly less correlated to the blue and red values of the neighbouring pixel.
- The images tend to be symmetric about the vertical and horizontal. For example, the colour of the top-left pixel is correlated to the colour of the top-right pixel and the bottom-left pixel. This kind of symmetry can be observed in all pixels in the faint anti-diagonals of the covariance matrix. It is probably caused by the way people take photographs – making the ground plane horizontal and centering on the main object.

Chapter 3

Object classification experiments

3.1 The labeled subset[1]

We paid students to label a subset of the tiny images dataset. The labeled subset we collected consists of ten classes of objects with 6000 images in each class. The classes are *airplane*, *automobile* (but not truck or pickup truck), *bird*, *cat*, *deer*, *dog*, *frog*, *horse*, *ship*, and *truck* (but not pickup truck). Since each image in the dataset already comes with a noisy label (the search term used to find the image), all we needed the labelers to do was to filter out the mislabeled images. To that end, each labeler was given a class and asked to examine all the images which were found with that class as the search term. Since the dataset contains only about 3000 images per search term, the labelers were also asked to examine all the images which were found with a search term that is a hyponym (as defined by WordNet) of the main search term. As an example, some of the hyponyms of ship are cargo ship, ocean liner, and frigate. The labelers were instructed to reject images which did not belong to their assigned class. The criteria for deciding whether an image belongs to a class were as follows[2]:

- The class name should be high on the list of likely answers to the question "What is in this picture?"
- The image should be photo-realistic. Labelers were instructed to reject line drawings.
- The image should contain only one prominent instance of the object to which the class refers.
- The object may be partially occluded or seen from an unusual viewpoint as long as its identity is still clear to the labeler.

The labelers were paid a fixed sum per hour spent labeling, so there was no incentive to rush through the task. Furthermore, we personally verified every label submitted by the labelers. We removed duplicate images from the dataset by comparing images with the L2 norm and using a rejection threshold liberal enough to capture not just perfect duplicates but also re-saved JPEGs and the like. Finally, we divided the dataset into a training and test set, the training set receiving a randomly-selected 5000 images from each class.

We call this the CIFAR-10 dataset, after the Canadian Institute for Advanced Research, which funded the project. In addition to this dataset, we have collected another set – 600 images in each of 100 classes. This we call the CIFAR-100 dataset. The methodology for collecting this dataset was identical to that for CIFAR-10. The CIFAR-100 classes are mutually exclusive with the CIFAR-10 classes, and so they can be used as negative examples for CIFAR-10. For example, CIFAR-10 has the classes *automobile* and *truck*, but neither of these classes includes images of pickup trucks. CIFAR-100 has the class *pickup truck*. Furthermore, the CIFAR-100 classes come in 20 superclasses of five classes each. For example, the superclass *reptile* consists of the five classes *crocodile*, *dinosaur*, *lizard*, *turtle*, and *snake*. The idea is that classes within the same superclass are similar and thus harder to distinguish than classes that belong to different superclasses. Appendix D lists the entire class structure of CIFAR-100.

All of our experiments were with the CIFAR-10 dataset because the CIFAR-100 was only very recently completed.

[1] We thank Vinod Nair for his substantial contribution to the labeling project.

[2] The instruction sheet which was handed out to the labelers is reproduced in Appendix C.